

# ccFEM User Manual

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## 1. Introduction

To address the issue of large errors in classical solid mechanics solvers when dealing with highly non-uniform deformations, ccFEM is developed from a novel higher-order deformation theory based on the concept of Homogenized Energy Density. Based on a fully coupled algorithm, it simultaneously solves the displacement field, strain field, and damage field. Compared with classical solid mechanics theory and gradient theory, this theory can simultaneously describe both hardening and softening effects caused by microscopic non-uniform deformation, and requires only the introduction of a single scale parameter with clear physical meaning and easy calibration, namely the Representative Volume Element (RVE) size. ccFEM employs second-order elements for both displacement and strain fields simultaneously, with solution efficiency comparable to classical theory-based solvers. Therefore, ccFEM is more suitable for solving complex deformation problems that cannot be accurately described by classical theory, such as:

- Problems regarding the size effect;
- Problems with highly concentrated stress/strain (or singularities);
- Mechanical behavior of complex heterogeneous materials (such as composites, meta-materials, thin-film materials, coatings, etc.);
- Damage and fracture problems simultaneously considering size effects...

For related theories, solution algorithms, and experimental verification, please refer to:

1. Cao Y.H., Zhang C.Y.\*, Finite-deformation Damaged Homogenized Energy Model for Quasi-brittle Fracture, Engineering Fracture Mechanics, 2026, under revision.
2. Cao Y.H., Zhang C.Y.\*, Damaged Homogenized Energy Model based on a Decomposition of Higher-Order Strain Energy Density, International Journal of Engineering Science 223(2026), 104498.
3. Cao Y.H., Zhang C.Y.\*, A Two-scale High-order Damaged Elasticity Theory and Solution Procedure for Quasi-brittle Fracture, Computer Methods in Applied Mechanics and Engineering 445 (2025), 118206.
4. Sang M.S., Zhang C.Y.\*, Cao Y.H., Influence of size and shape of defects on deformation concentration and brittle fracture of perforated plates, Journal of Engineering Mechanics 151(2025), 04024117.
5. Cao Y.H., Zhang C.Y.\*, Wang B., A New high-order deformation theory and solution procedure based on homogenized strain energy density, International Journal of Engineering Science 195 (2024), 103990.

6. Sang M.S., Zhang C.Y.\*, Cao Y.H., Wang B., Experimental and theoretical evaluation of influence of hole size on deformation and fracture of elastic perforated plate, *Mechanics of Materials* 187(2023), 104841.

## 2. Runtime Environment

ccFEM contains only the solver. Input files are defined in text format (see Section 5.1 Input File for details). It is recommended to use **Gmsh** to generate meshes (see Section 4 Preprocessing Methods) and **Paraview** for post-processing.

Currently, it only runs on Linux systems. On Windows systems, the program can be run through WSL or a Linux virtual machine.

For ease of use, the development team has packaged the ccFEM executable and runtime environment using Docker. On a Linux system with Docker installed, download ccFEM using the following command:

```
docker pull acculix/solvers:latest
```

If you encounter network connection issues, it is recommended to use a domestic mirror:

```
docker pull docker.1ms.run/acculix/solvers:latest
```

## 3. Quick Start

First, confirm that Docker is installed on your Linux or WSL system.

| Step | Command                                                                                                                                                                                                                                                      | Remarks                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 1    | docker pull acculix/solvers:latest<br><br>[If network issues occur, use domestic mirror:<br>docker pull docker.1ms.run/acculix/solvers:latest]                                                                                                               | Download ccFEM solver environment                            |
| 2    | mkdir work<br>cd work                                                                                                                                                                                                                                        | Create and enter working directory                           |
| 3    | docker run -it --rm -v \$(pwd):/work -w /work<br>acculix/solvers:latest /bin/bash<br><br>[If using domestic mirror:<br>docker run -it --rm -v \$(pwd):/work -w /work<br>docker.1ms.run/acculix/solvers:latest /bin/bash]<br><br>If normal, type exit to quit | Check if ccFEM image is normal                               |
| 4    | Create run.sh script in work directory with:<br>#!/bin/sh<br>sudo docker run -it --rm -v \$(pwd):/work -w /work<br>acculix/solvers:latest /bin/bash "\$@"<br><br>Then add execution permission:<br>chmod +x ./run.sh                                         | For easy job submission                                      |
| 5    | Use Gmsh to generate mesh file, write input file model.txt, and submit calculation:                                                                                                                                                                          | model.txt completely defines mesh file, material parameters, |

|   |                                                                                 |                                                 |
|---|---------------------------------------------------------------------------------|-------------------------------------------------|
|   | <code>./run.sh mpirun --allow-run-as-root -n 2<br/>ccFEM_mpi ./model.txt</code> | boundary conditions, and<br>solution parameters |
| 6 | Use Paraview to open calculation results                                        |                                                 |

## 4. Preprocessing Method

This section recommends using Gmsh ([www.gmsh.info/](http://www.gmsh.info/)) to establish geometric models and generate meshes.

Gmsh is an open-source preprocessing tool and an excellent open-source mesh generation software. It mainly contains four modules: Geometry, Mesh, Solver, and Post-processing. The **Geometry module** uses Boundary Representation (BRep) to define geometric information. The main geometric entities include points, lines, surfaces, volumes, and other common elements in 3D space. In addition to geometric entities, Gmsh also supports defining **Physical Groups**. Physical Groups logically classify different geometric entities into one category, which is very convenient when setting material properties and boundary conditions. The **Mesh module** provides multiple algorithms to discretize geometry and generate high-quality meshes. Gmsh's built-in algorithms include the Frontal method, Delaunay, and Adaptive methods. Without specification, Gmsh will "automatically" select the most suitable algorithm based on geometric model information. The following introduces the method of generating meshes with Gmsh.

### 4.1 Geometric Modeling

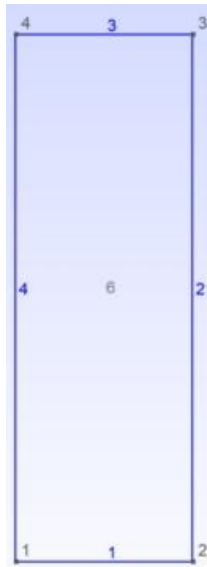
Gmsh supports both **interactive modeling** and **script modeling**. Since parameterization can be performed through scripts, for problems with complex geometric structures, **script modeling is preferred**.

In the script modeling process, there are four important types of commands: defining parameters, points, lines, and surfaces. A simple example is as follows:

```
// Comments use "//"
lc = 0.009; // Characteristic length parameter, defines element size around
points
Point(1) = {0, 0, 0, lc}; // Point definition uses (x,y,z) and characteristic
length lc
Point(2) = {.1, 0, 0, lc}; // All data can be defined as variables
Point(3) = {.1, .3, 0, lc};
Point(4) = {0, .3, 0, lc};
Line(1) = {1,2}; // Straight line defined by a series of points, line 1 starts
at point 1, ends at point 2
Line(2) = {3,2};
Line(3) = {3,4};
Line(4) = {4,1};
// To define a surface formed by 4 lines, need to create line loop first
// Line loop is a series of connected lines, sign of each line indicates
direction
Line Loop(5) = {4,1,-2,3}; // -2 means line(2) direction is opposite to defined
direction
Plane Surface(6) = {5}; // Define surface with line loop, since no holes, only
one line loop here
// At this point Gmsh can display surface 6 and mesh it
```

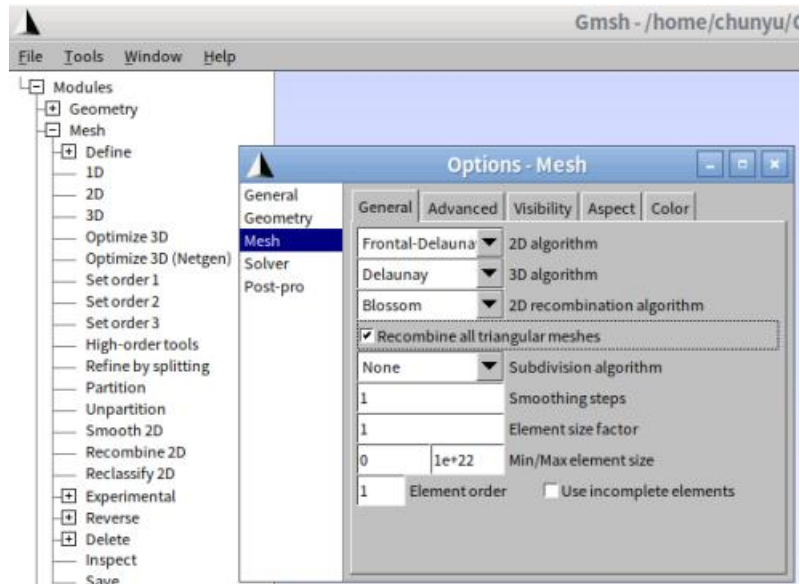
```
// If you want to assign region to different mesh units, need to define
Physical Groups
// Physical Groups combine different geometric entities through common region
numbers
Physical Point(1) = {1,2}; // Combine points 1,2 in physical group 1, physical
group ID is 1
MyLine = 99;
Physical Line(MyLine) = {1,2,4}; // Combine lines 1,2,4, physical group ID is
99
Physical Surface("My fancy surface label") = {6}; // All elements in surface
automatically assigned region number 100
```

Save the above content as a text file with .geo extension. Gmsh reads and displays it as follows:

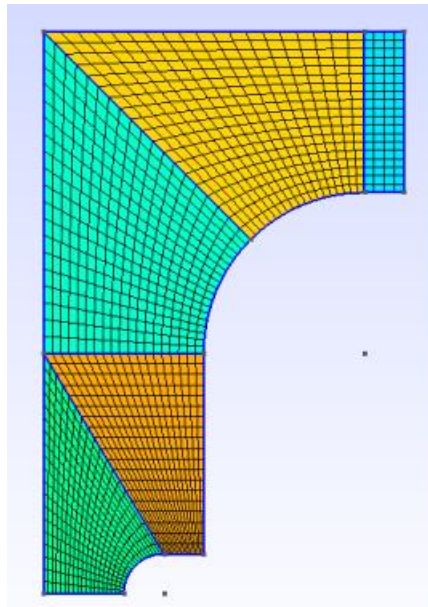


## 4.2 Mesh Generation

ccFEM recommends using quadrilateral meshes (for 2D problems) and hexahedral meshes (for 3D problems). To generate quadrilateral elements, you need to check "Recombine all triangular meshes" via Tools→Options→Mesh→General.



Then select Mesh→2D (3D) from the left tree view.



Finally, save the generated mesh through File→Save Mesh, with extension .msh.

## 5. Solution

### 5.1 Input File

The input file content includes: mesh file, material properties, boundary conditions, and output parameters, etc. A typical input file is shown below:

```
modelspace = planestress #planestress, planestrain or 3D
```

```

[mesh]
    meshfile = 'your mesh file' # Read the specified mesh file
    order = 2 #FE order. may not be equal to mesh order
    meshrefinement = 0 #number of times for uniform mesh refinement

[materials] #Material Properties
    [./anymaterial]
        E = 200e3 # Elastic modulus
        v = 0.3 # Poisson's ratio
        lengthscale = 0.6 # RVE size
        MaxStrain = 0.00018 #Critical strain, beyond which damage evolves,
        #if MaxStrain=0 damage never occurs.
        DamageBeta = 1765 #Parameter, control speed of damage evolution
    []

[region_properties]
    [./wholeregion]
        elementsetid = 0 #start from 0
        material = anymaterial
    []

[boundaries] # boundary conditions
    [./botEdge]
        elementsetid = 0 #start from 0
        type = displacement # Boundary condition type
        uy = 0.0 #displacement or traction
    [../topEdge]
        elementsetid = 1
        type = traction
        ty = 1.0
    [../leftEdge]
        elementsetid = 3
        type = displacement
        ux = 0.0
    []

[solution] # options for solution
    thermalmechanical = false
    finitedeformation = false
    admm = false #for 3D problems, use admm to avoid ill
    #sparse stiffness matrix
    newton = true #control explicit/implicit solution of
    #geometry nonlinearity and damage
    reorderingreuse = true #if mumps solver give errors with stiffness
    #data, set "false" to avoid errors, but
    #solution slows down
    skipzeros = 1 #if mesh quality is too poor to solve, set
    #"0", which makes solution slow down
    totaltime = 1.0e1 #real time for thermal solution
    penalty = 1.0e5
    linearsolver = MUMPS #now always use mumps solver, to be
    #extended to others solvers like PCG
    tolerance = 1.e-6
    maximumiteration = 2000 #max step for admm iterations
    printlevel = 0

```

```

damagemaximumiteration = 200 # max step for damage iterations
maximumstepsize = 0.05 #if <0.002, explicit solution will be applied.

[postprocess]
    paraview = 1

```

The parameter descriptions for each part of the input file are shown in the table below.

**Table 1. Input File Parameter Description**

| Parameter           |                       | Sub-parameter        | Description                                                                                                                                                                                                                                                                                   |
|---------------------|-----------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| modelspace          |                       |                      | planestress, planestrain or 3D                                                                                                                                                                                                                                                                |
| [mesh]              |                       | meshfile             | Read specified mesh file                                                                                                                                                                                                                                                                      |
|                     |                       | order                | Finite element discretization order ( $\geq 2$ ), can differ from element order, default is 2                                                                                                                                                                                                 |
|                     |                       | meshrefinement       | Number of mesh refinements, default is 0                                                                                                                                                                                                                                                      |
| [materials]         | Mechanical Parameters | E                    | Elastic modulus                                                                                                                                                                                                                                                                               |
|                     |                       | v                    | Poisson's ratio                                                                                                                                                                                                                                                                               |
|                     |                       | lengthscale          | Scale parameter (RVE size)                                                                                                                                                                                                                                                                    |
|                     |                       | MaxStrain            | Critical strain, damage evolves beyond this; if MaxStrain=0, damage never occurs                                                                                                                                                                                                              |
|                     |                       | DamageBeta           | Larger value means faster damage evolution from 0 to 1                                                                                                                                                                                                                                        |
|                     |                       | MaterialDamage       | When value > 0, entire material region becomes initial crack                                                                                                                                                                                                                                  |
|                     | Thermal Parameters    | CTE                  | Thermal expansion coefficient                                                                                                                                                                                                                                                                 |
|                     |                       | k                    | Thermal conductivity, unit mJ/mm/K                                                                                                                                                                                                                                                            |
|                     |                       | Q_hs                 | Internal heat source, unit mW/mm <sup>3</sup>                                                                                                                                                                                                                                                 |
|                     |                       | rho                  | Density, t/mm <sup>3</sup>                                                                                                                                                                                                                                                                    |
|                     |                       | T_ref                | Ambient temperature                                                                                                                                                                                                                                                                           |
|                     |                       | T_init               | Initial temperature                                                                                                                                                                                                                                                                           |
|                     |                       | C_p                  | Specific heat, unit mJ/t/K                                                                                                                                                                                                                                                                    |
| [region_properties] |                       | elementsetid         | Physical group ID in this region, starting from 0                                                                                                                                                                                                                                             |
|                     |                       | material             | Material name for this region                                                                                                                                                                                                                                                                 |
| [BCs]               |                       | elementsetid         | Physical group ID for boundary, starting from 0                                                                                                                                                                                                                                               |
|                     |                       | type                 | displacement (prescribed displacement) or traction (surface force)                                                                                                                                                                                                                            |
|                     |                       | ux/uy/uz or tx/ty/tz | Specified value. Supports function definition                                                                                                                                                                                                                                                 |
| [solution]          |                       | penalty              | Penalty factor, same order of magnitude as E                                                                                                                                                                                                                                                  |
|                     |                       | linearsolver         | Linear equation solver, currently only MUMPS is supported                                                                                                                                                                                                                                     |
|                     |                       | thermalmechanical    | Enable thermo-mechanical coupling; error if thermal parameters undefined                                                                                                                                                                                                                      |
|                     |                       | finitedeformation    | Enable finite deformation solution                                                                                                                                                                                                                                                            |
|                     |                       | admm                 | Enable alternating direction coupling for high-order deformation; enters iterative state. If considering damage or geometric nonlinearity, updates damage evolution and solves geometric nonlinearity simultaneously during iteration. Explicit/implicit update influenced by newton setting. |

|               |  |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------|--|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |  | newton                 | Enable implicit iteration for damage and geometric nonlinearity; disable for explicit solution. When enabled, for finite deformation with damage problems, if continuously non-convergent, geometric nonlinearity uses explicit update while damage remains implicit; if damage iteration still non-convergent, both use explicit solution.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|               |  | reorderingreuse        | Default enabled. After each load/time step, only factorize stiffness matrix once to solve inverse; each implicit iteration or admm iteration directly solves stiffness matrix inverse. If mesh quality is poor causing mumps sparse matrix errors, disable this option, re-factorize each iteration, significantly slower.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|               |  | skipzeros              | Default is 1, computed element stiffness matrix 0 elements occupy memory in overall sparse matrix assembly. If poor mesh quality or severe damage causes local element singularity and errors, set to 0 (note this is not bool), increases memory usage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|               |  | maximumiteration       | When admm is enabled, affects total admm iteration count (damage and geometric nonlinearity updated simultaneously during admm iteration)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|               |  | damagemaximumiteration | When admm is not enabled, affects damage/geometric nonlinearity iteration count. Generally damagemaximumiteration << maximumiteration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| [postprocess] |  | paraview               | Whether to output paraview format results, 1=output, 0=no output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|               |  | outputs                | <p>Output results. Default outputs paraview results.</p> <p>EquiStrain/EquiStress: Equivalent strain/stress affecting damage evolution, only contributed by pure tensile strain;</p> <p>LoadForce/SupportReaction: Load force/support reaction at each displacement loading end. Automatically outputs individual forces when multiple fixed/loading ends exist, and total LoadForce_tot when multiple loading ends.</p> <p>Physical quantities with Degraded prefix represent degraded physical quantities considering damage, removing physical effects of damaged regions; when damage occurs, only focus on Degraded prefix quantities.</p> <p>Engineering strain: When finite deformation is enabled, other strain physical quantities are based on Hencky strain; only this quantity is engineering strain, automatically considering damage degradation.</p> <p>Other unmentioned physical quantities are model test reference quantities, not for reference.</p> |

## 5.2 Program Execution

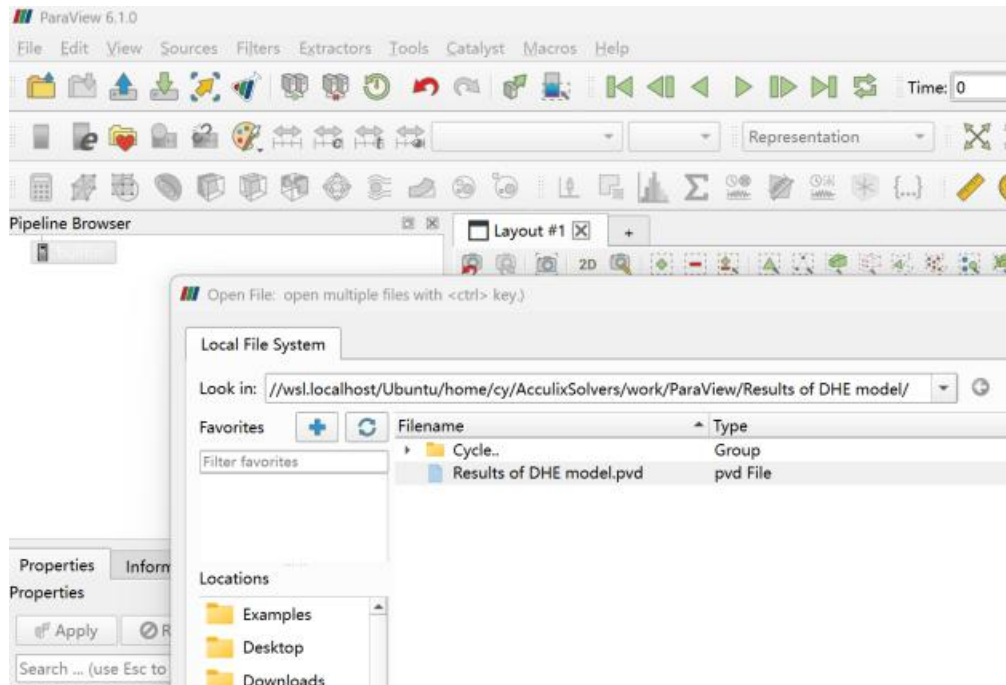
Prepare the input file in the above format and save it, then enter in the terminal: `mpirun -n 4 ccFEM_mpi ./yourinput.txt`, where `yourinput.txt` is the prepared input file. The solver reads all settings and starts calculation. The normal running screen is shown below:

```
dim(L) = 21465
dim(U+Ud+L) = 57240
*****
leftsupportEnd is set as point constraint.
ux of boundary leftsupportEnd is set to be 0.
uy of boundary leftsupportEnd is set to be 0.
rightsupportEnd is set as point constraint.
uy of boundary rightsupportEnd is set to be 0.
px of boundary loadingEnd1 is set to be 60.
py of boundary loadingEnd1 is set to be 10.
uy of boundary loadingEnd1 is set to be -30.
px of boundary loadingEnd2 is set to be 160.
py of boundary loadingEnd2 is set to be 10.
uy of boundary loadingEnd2 is set to be -30.
Solution begins...
Physical fields : LoadForceTOT LoadForce1 SupportReactionTOT LoadForce2 Supp
ement multiplier damage degraded_mises SEDplus degradedSED strain averaged
ded_strain EquiStrain engineering_strain will be output.
Incremental solution begins:
Damage is considered...
Small deform trial step:
Incremental step 1 (with 0.000% + 1.000% Loading) begins...
1      |ε-ε_old|/|ε| = 1.0000000
2      |ε-ε_old|/|ε| = 0.0000000
Incremental step 1 (with 1.000% Loading) finished.
physical fields are output in Paraview at 0.010
physical fields are output in Glvis at 0.010
=====
Trial step finished
=====
```

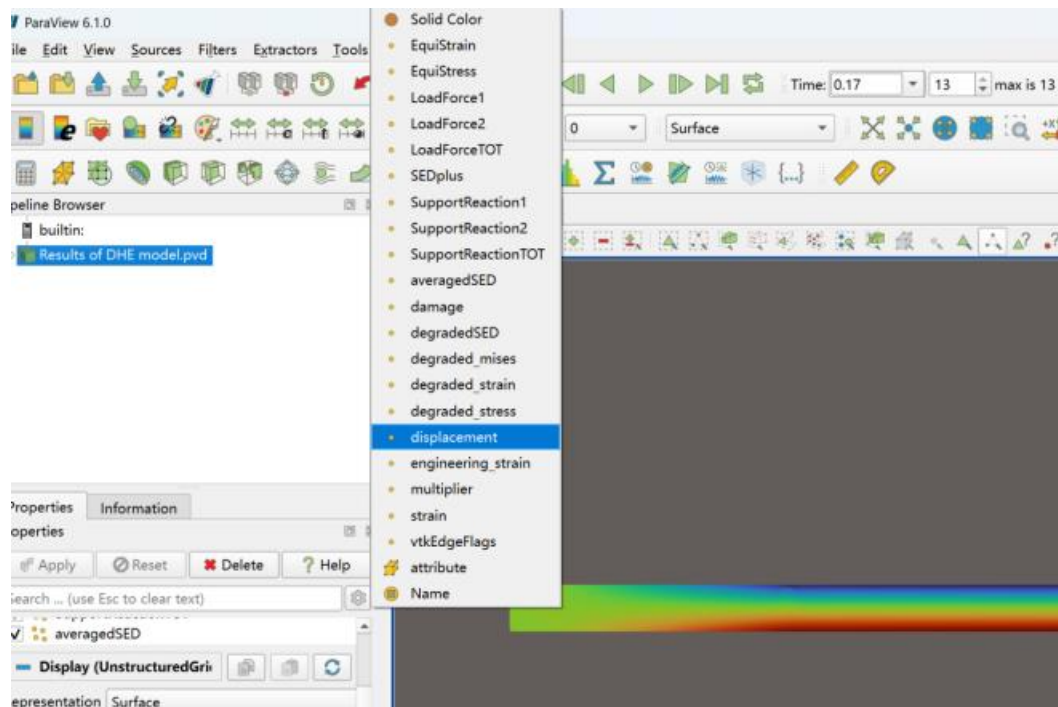
## 5.3 Post-processing Result Analysis

It is recommended to output Paraview format results (need to specify `paraview=1` in the [postprocess] section of the input file) and use Paraview for result viewing.

If the program runs without error, it will create a Paraview directory and save Paraview format result files to this directory. Run the Paraview program to import result files for viewing and analysis:



Select output results to view contour:



The

meanings of each field variable are shown in Table 2:

**Table 2. Field Variable Meaning Description**

| Field Variable | Meaning                                                                       | Remarks                                                                                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| strain         | Small deformation: infinitesimal strain;<br>Finite deformation: Hencky strain | 2D component order: $\epsilon_{11}$ , $\epsilon_{22}$ , $\epsilon_{12}$<br>3D component order: $\epsilon_{11}$ , $\epsilon_{22}$ , $\epsilon_{33}$ , $\epsilon_{12}$ ,<br>$\epsilon_{13}$ , $\epsilon_{23}$ |

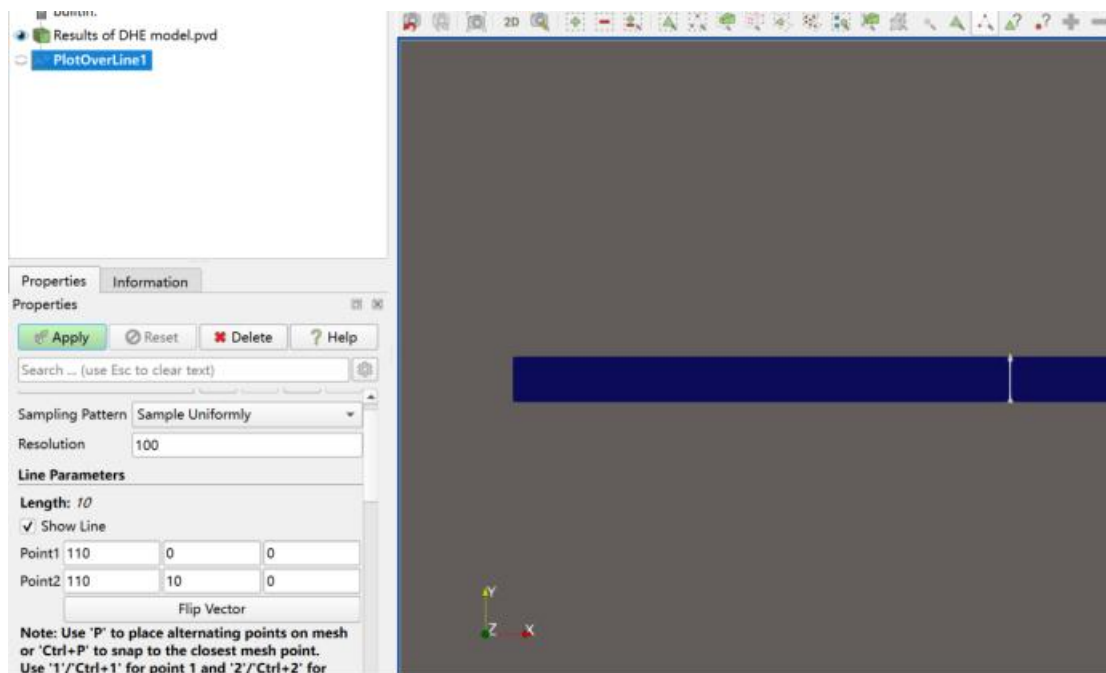
|                         |                                                                                                    |                                                                                                                                   |
|-------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Engineering_strain      | Degraded engineering strain (only differs from Degraded_strain when finite deformation is enabled) | Component order same as strain                                                                                                    |
| Degraded_strain         | Degraded strain after damage occurs, $(1-d)\epsilon$                                               | Component order same as strain; when no damage, Degraded_strain equals strain                                                     |
| Stress, Degraded_stress | Stress, degraded stress                                                                            | Component order same as strain; when no damage, Degraded_stress equals stress                                                     |
| Damage                  | Damage factor                                                                                      | $[0,1]$ , 0=no damage, 1=complete damage (crack)                                                                                  |
| LoadForce               | External force at each non-zero displacement loading end                                           | Automatically outputs individual forces for each loading end, and total LoadForce_tot when multiple loading ends exist            |
| SupportReaction         | Reaction force at each fixed end                                                                   | Automatically outputs individual reaction forces for each fixed end, and total SupportReaction_tot when multiple fixed ends exist |
| Degraded_mises          | Degraded Mises stress                                                                              |                                                                                                                                   |
| EquiStress/EquiStrain   | Degraded equivalent stress/strain                                                                  |                                                                                                                                   |
| DegradedSED             | Degraded high-order strain energy density                                                          | When no damage, represents high-order SED                                                                                         |
| SEDplus                 | Degraded positive strain energy density $(1-d)U^+$ (without high-order terms)                      | When no damage, represents $U^+$                                                                                                  |

## 5.4 Common Post-processing Techniques

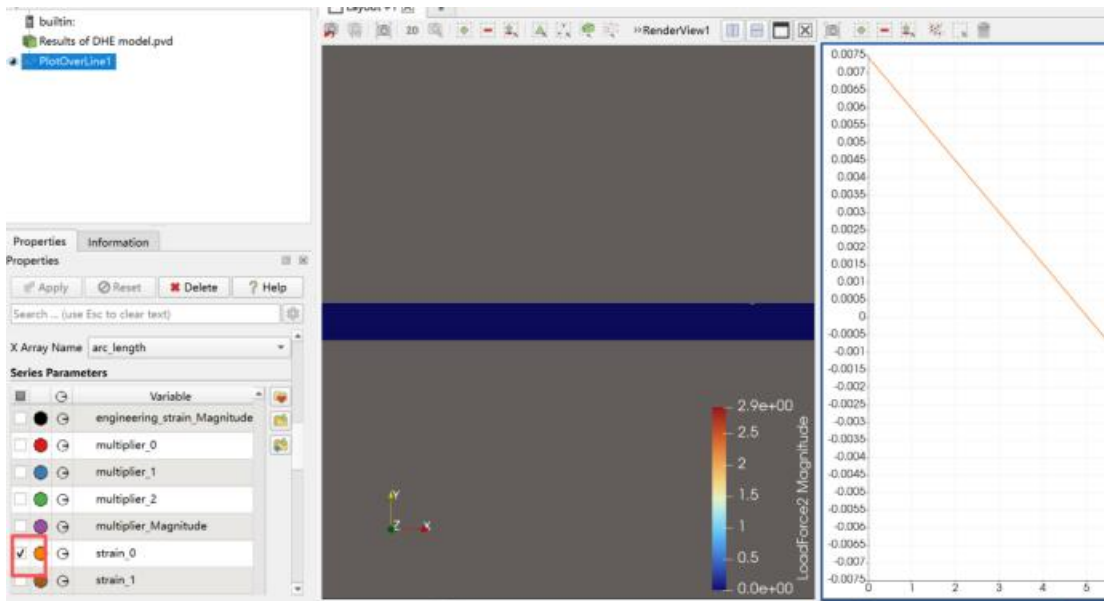
### 5.4.1 Distribution along Structural Line

Menu: Filters→Data Analysis (or Domain→Data Analysis)→Plot Over Line

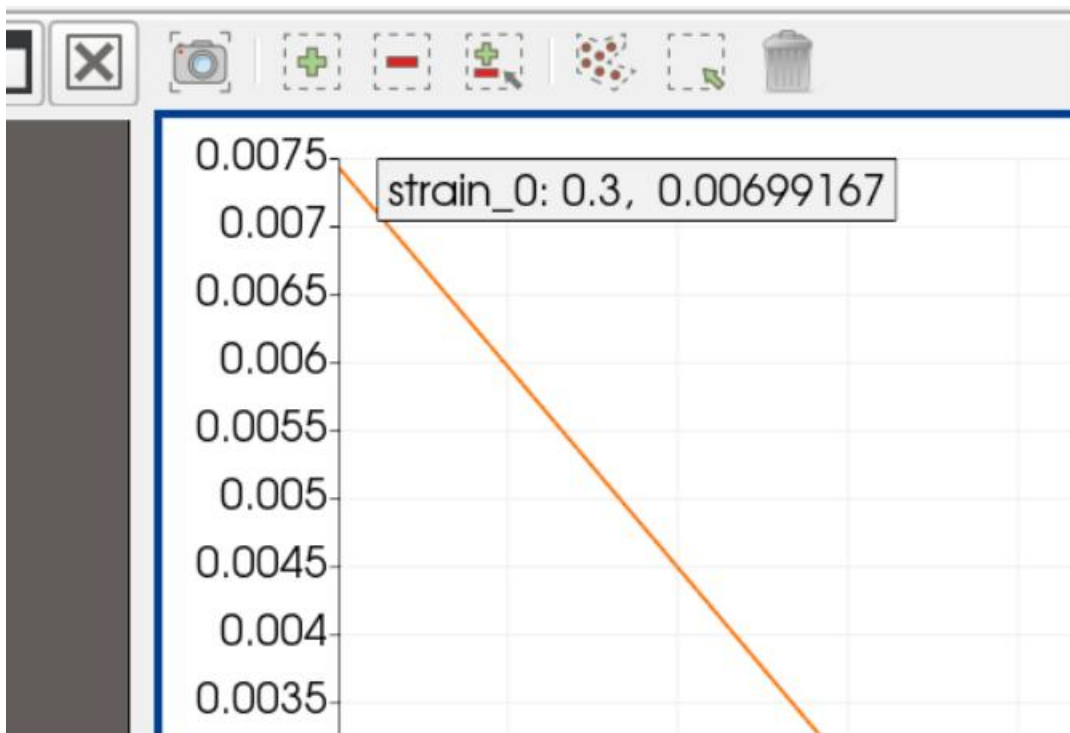
Enter the start and end point coordinates of the line and click "Apply":



Select the field results to analyze:

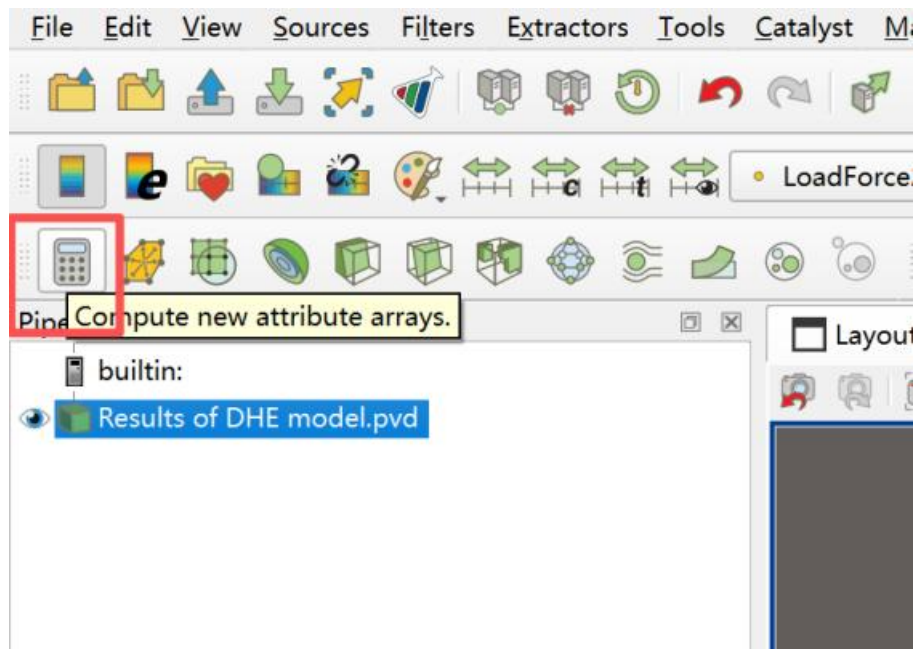


Move the mouse to the data on the right to view specific position results:

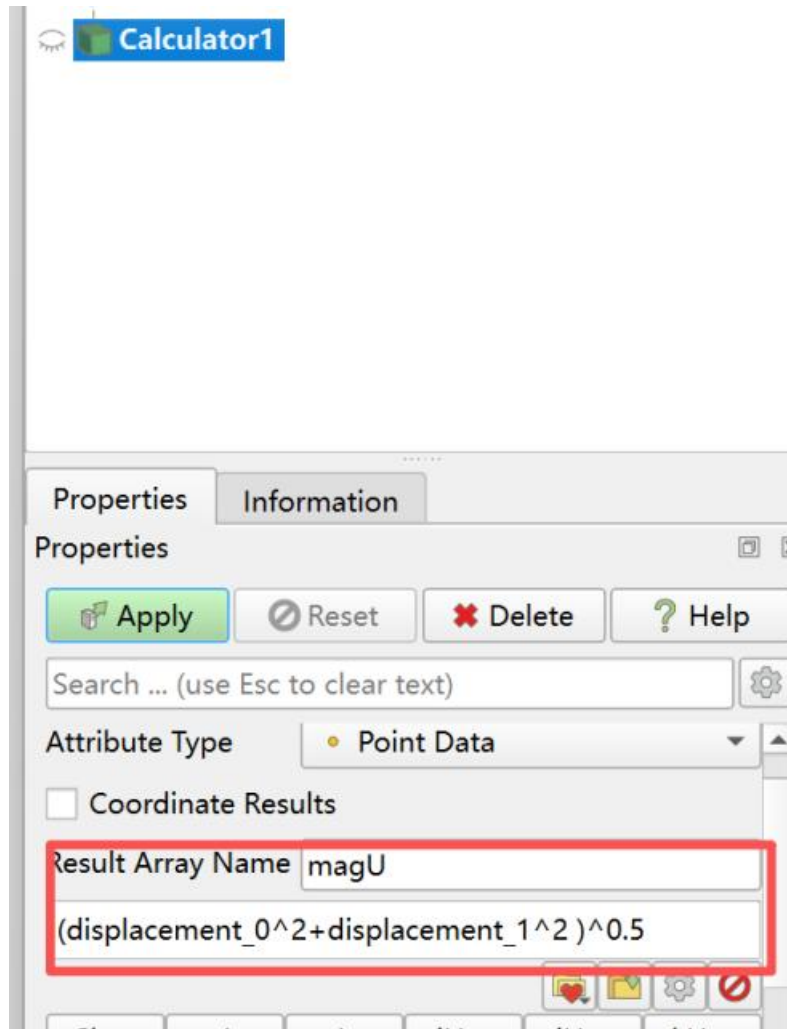


#### 5.4.2 Operations on Existing Fields

Select Calculator:



Define field results and generation method:



Click "Apply" to view results.

## 6. Typical Examples

### 6.1 Perforated Plate Tension — Stress/Strain Concentration Dependence on Hole Radius

**Gmsh modeling file:**

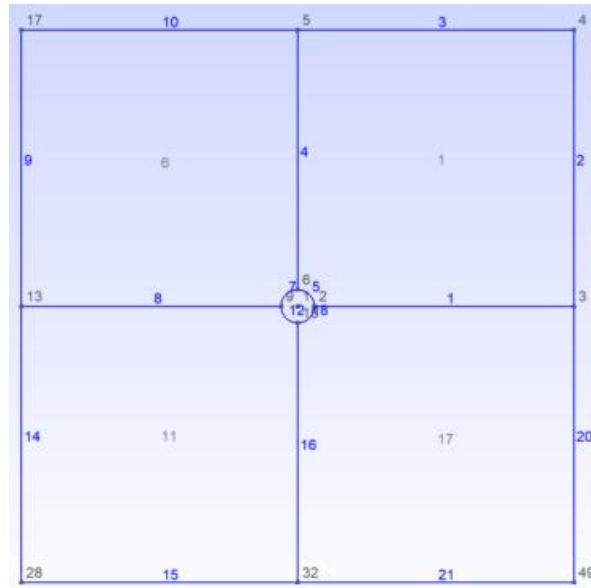
```
r=0.9; // Define circular hole diameter
l=15; // Define distance from center of square plate to edge
N=10;
Point(1) = {0, 0, 0, 1.0}; // Point
Point(2) = {r, 0, 0, 1.0};
Point(3) = {l, 0, 0, 1.0};
Point(4) = {l, l, 0, 1.0};
Point(5) = {0, l, 0, 1.0};
Point(6) = {0, r, 0, 1.0};
Line(1) = {2, 3}; // Generate lines from above points
```

```

Line(2) = {3, 4};
Line(3) = {4, 5};
Line(4) = {5, 6};
Circle(5) = {6, 1, 2};
Curve Loop(1) = {5, 1, 2, 3, 4}; // 5, 1, 2, 3 and 4 form closed curve surface
Plane Surface(1) = {1}; // Form new surface
Symmetry {1, 0, 0, 0} {
    Duplicata { Surface{1}; }
}
Symmetry {0, 1, 0, 0} {
    Duplicata { Surface{6}; Surface{1}; }
}
Physical Surface(1) = {1,6,11,17}; // Define a physical surface, save 1, 6, 11
and 17 as same region number
Physical Curve(1) = {15,21}; // Define a physical curve, save 15 and 21 as same
region number
Physical Curve(2) = {10,3};
Physical Curve(3) = {5,7,12,18};
Physical Curve(4) = {9,14};
Transfinite Curve{2} = N; // Use N=10 nodes on curve 2, structured mesh
algorithm, right side is number of nodes on line, evenly distributed
Transfinite Curve{3} = N;
Transfinite Curve{5} = 2*N-1; // Use 19 nodes on curve 5
Transfinite Curve{1,-4} = 3*N Using Progression 1.1; // Place 3*10 nodes on
lines 1 and 4, line 4 reversed, Using Progression 1.1 means each element width
is 1.1 times the previous
Transfinite Surface{1} = {2, 3, 5, 6}; // Specify 4 corner points (node IDs)
for meshing surface
Transfinite Curve{10} = N;
Transfinite Curve{9} = N;
Transfinite Curve{7} = 2*N-1;
Transfinite Curve{8,-4} = 3*N Using Progression 1.1;
Transfinite Surface{6} = {6,5,13,9};
Transfinite Curve{14} = N;
Transfinite Curve{15} = N;
Transfinite Curve{12} = 2*N-1;
Transfinite Curve{8,-16} = 3*N Using Progression 1.1;
Transfinite Surface{11} = {9,13,32,18};
Transfinite Curve{20} = N;
Transfinite Curve{21} = N;
Transfinite Curve{18} = 2*N-1;
Transfinite Curve{1,-16} = 3*N Using Progression 1.1;
Transfinite Surface{17} = {2,18,32,3};
Mesh.MshFileVersion = 2.2; // Output in 2.2 format

```

The specific model is shown in the figure below:



### Input file:

```

modelspace = planestress #planestress, planestrain or 3D

[mesh]
meshfile = './mesh_mat.msh' #./data/wholesquare_hole_2d.msh
order = 2 #FE order. may not be equal to mesh order
meshrefinement = 0 #number of times for uniform mesh refinement

[materials]
[./anymaterial]
E = 4.0e3
v = 0.4
lengthscale = 1.1
# [./perturbation]
# randomField = './randomField.gf'
# scale = 0.02 #[-1,1]*scale
[]

[region_properties]
[./wholeregion]
elementsetid = 0 #start from 0
material = anymaterial
[]

[boundaries] # boundary conditions
[./botEdge]
elementsetid = 0 #start from 0
type = displacement
#ux = 0.0
uy = 0.0
[./topEdge]
elementsetid = 1
type = traction
#uy = 1.0

```

```
ty = 1.0
[../leftEdge]
elementsetid = 3
type = displacement
ux = 0.0
[]

[solution] # options for solution
penalty = 1.0e6
linearsolver = umfpack

[postprocess]
paraview = 1
output = 'u,e,s1,sec' # u-displacement, e-corrected strain, s0-stress from
displacement (conventional stress), s1-stress from corrected strain, se-strain
energy
```